

Insup Lee

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Research Interests

My research lies at the intersection of security and AI systems. I currently work on accelerating robot demonstrations for faster task execution with VLA policies, and aim to improve the safety and trustworthiness of physical AI.

Education

Korea University	Seoul, Korea
Ph.D. in Cybersecurity (Advisors: Sangjin Lee and Seokhie Hong)	Sep 2019 - Aug 2026
Korea University	Seoul, Korea
B.E. in Cyber Defense	Mar 2014 - Feb 2018

Work Experience

KAIST AI	Seoul, Korea
Postdoctoral Researcher, LLM Innovation Research Center (Advisor: Dongsu Han)	Sep 2026 - Present
Ministry of National Defense	Seoul, Korea
Cybersecurity Officer (Army Captain), Republic of Korea Army	Aug 2023 - May 2025
Agency for Defense Development (ADD)	Seoul, Korea
Researcher, Cyber Technology Center (Mentor: Changhee Choi)	Jul 2018 - Jul 2023

Publications

(C: conference, J: journal, P: preprint, †: corresponding author)

International Publications

- C5 *Schema-Agnostic Knowledge Graph Construction via Hybrid Ontology Discovery for Cyber Threat Intelligence*
S. Kim, J. Kim, D. Kang, D. Kim, and **I. Lee**[†]
Annual Computer Security Applications Conference (**ACSAC'26**), Dec. 2026.
- C4 *Mnemonic Hack: Recovering the Master Seed from Bitcoin Hardware Wallets via Side-Channel Attacks*
J. Baek, D. Bae, G. Kim, G. Ahn, **I. Lee**, S. Park, S. Hong, and H. Kim
ACM Conference on Computer and Communications Security (**CCS'26**), Nov. 2026.
- J10 *Multi-Domain Side-Channel Analysis for Anomaly Detection in Embedded Systems*
S. Park, D. Bae, **I. Lee**, J. Kim, H. Oh, S. Hong, and H. Kim
IEEE Embedded Systems Letters, 2026.
- C3 *Exploiting Per-Core Leakage: Electromagnetic Side-Channel Monitoring of Multicore Architectures*
D. Bae, S. Park, **I. Lee**, Y. Jung, K. Lee, H. Kim, and S. Hong
ACM/IEEE Design Automation Conference (**DAC'26**), Jul. 2026.
- J9 *MuCamp2: Generating Validated Cyber Campaign Variants via Constrained LLMs for Group Attribution*
K. Kim, J. Lee, **I. Lee**, I. Han, J. Lee, S. Shim, and C. Choi
IEEE Access, Vol. 14, pp. 87329-87344, Jun. 2026.
- J8 *LeakDiT: Diffusion Transformers for Trace-Augmented Side-Channel Analysis*
I. Lee, D. Bae, S. Hong, and S. Lee
IEEE Computer Architecture Letters, Vol. 25, No. 1, pp. 5-8, Jan./Jun. 2026.
- J7 *Multi-Step LLM Pipeline for Enhancing TTP Extraction in Cyber Threat Intelligence*
H. Kim, D. Lee, **I. Lee**, S. Lee, and S. Lee
IEEE Access, Vol. 13, pp. 179696-179710, Oct. 2025.

- J6 *Enhancing Modulation Classification via Diffusion Transformers for Drone Video Signal Processing*
I. Lee[†], K. Alteneiji, and M. Alghfeli
IEEE Signal Processing Letters, Vol. 32, pp. 3325-3329, Aug. 2025.
- J5 *MuCamp: Generating Cyber Campaign Variants via TTP Synonym Replacement for Group Attribution*
I. Lee and C. Choi
IEEE Transactions on Information Forensics and Security (TIFS), Vol. 20, pp. 6162-6174, Jun. 2025.
- J4 *UniQGAN: Towards Improved Modulation Classification With Adversarial Robustness Using Scalable Generator Design*
I. Lee and W. Lee
IEEE Transactions on Dependable and Secure Computing (TDSC), Vol. 21, No. 2, pp. 732-745, Mar./Apr. 2024.
- J3 *Camp2Vec: Embedding Cyber Campaign With ATT&CK Framework for Attack Group Analysis*
I. Lee and C. Choi
ICT Express, Vol. 9, No. 6, pp. 1065-1070, Dec. 2023.
- J2 *Exploiting TTP Co-occurrence via GloVe-Based Embedding With ATT&CK Framework*
C. Shin, **I. Lee**, and C. Choi
IEEE Access, Vol. 11, pp. 100823-100831, Sep. 2023.
- J1 *BAN: Predicting APT Attack Based on Bayesian Network With MITRE ATT&CK Framework*
Y. Kim, **I. Lee**, H. Kwon, G. Lee, and J. Yoon
IEEE Access, Vol. 11, pp. 91949-94968, Aug. 2023.
- C2 *Anomaly Dataset Augmentation Using Sequence Generative Models*
S. Shin, **I. Lee**, and C. Choi
IEEE International Conference on Machine Learning and Applications (ICMLA'19), Dec. 2019.
- C1 *Opcode Sequence Amplifier Using Sequence Generative Adversarial Networks*
C. Choi, S. Shin, and **I. Lee**
International Conference on ICT Convergence (ICTC'19), Oct. 2019.

Domestic Publications (Korean)

- D4 S. Park, W. Shin, D. Bae, H. Shin, **I. Lee**, S. Hong, and H. Kim, "Single Fault Injection and Countermeasure for MAVLink Authentication Enforcement in Unmanned Aerial Vehicles," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2026. (Selected as an Outstanding Paper Award)
- D3 S. Park, D. Bae, **I. Lee**, H. Kim, and S. Hong, "EM-Based Anomaly Detection using a Dual-Domain Approach," in Proc. of the KIISC Winter Conference (CISC-W), Nov. 2025. (Selected as an Outstanding Paper Award)
- D2 H. Park and **I. Lee**[†], "Enhanced DDoS Detection via Traffic Volume-Based Labeling and Transfer Learning," Journal of Internet Computing and Services (JICS), Vol. 26, No. 4, pp. 1-8, Aug. 2025.
- D1 K. Kim and **I. Lee**[†], "User Behavior Embedding via TF-IDF-BVC for Web Shell Detection," Journal of The Korea Institute of Information Security & Cryptology (JKIISC), Vol. 34, No. 6, pp. 1231-1238, Dec. 2024.

Other Experience

- AI Cyber Challenge (AIxCC), DARPA and ARPA-H, USA Apr 2024 – Aug 2024
- Participated in the semifinal round as a member of Team KORIA, submitting our cyber reasoning system that leverages LLMs for automated detection and patching of software vulnerabilities
- SW Outsourcing Development, KCMVP-Certified Cryptographic Module Jun 2017 – May 2018
- Implemented a cryptographic module with 25,000 LoC in C - ARIA block cipher (modes: ECB, CBC, CTR), hash functions (SHA-256, SHA-512), and HMAC-based DRBG for Windows (.dll) and Linux (.so)

Honors and Awards

- Outstanding Paper Award Nov 2026
CISC-W'26, KIISC
- KU Graduate School Achievement Award Feb 2026
Korea University, Seoul, Republic of Korea

- Outstanding Paper Award
CISC-W'25, KIISC Nov 2025
- Certificate of Commendation (UAE-ROK Engagement Program)
United Arab Emirates Ministry of Defense Mar 2025
- Ambassador's Commendation
Embassy of the Republic of Korea to the United Arab Emirates Mar 2025
- Full Tuition Scholarship (Korea University)
Ministry of National Defense, Republic of Korea Mar 2014 – Feb 2018

Academic Service

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC), 2025
- IEEE Transactions on Information Forensics and Security (TIFS), 2026
- IEEE/ACM Transactions on Networking (ToN), 2026
- IEEE Transactions on Communications (TCOM), 2025, 2026
- IEEE Journal on Selected Areas in Communications (JSAC), 2025, 2026

Teaching Experience

- Lecturer, Computer Networks (SCS302, Fall 2025), Korea University Sep 2025 - Feb 2026
- Instructor, Penetration Testing - Intermediate, UAE Ministry of Defense Sep 2024
- Instructor, Penetration Testing - Basic, UAE Ministry of Defense Jul 2024

Mentoring Experience

- **Seonwoo Kim** (Major at Ministry of National Defense) Feb 2026 – Present
Topic: LLM-based knowledge graph construction for CTI [C5]
- **Sujin Park** (Ph.D. Student at Korea University) Jun 2025 – Present
Topic: Side-channel analysis for anomaly detection [J10], [D3], [D4]
- **Hyoungrok Kim** (M.S. Student at Korea University - currently at XBOW) Mar 2025 – Sep 2025
Topic: LLM-based TTP extraction for CTI [J7]
- **Hyunjun Park** (Navy Lieutenant at Ministry of National Defense) Nov 2024 – Feb 2025
Topic: DDoS detection via transfer learning [D2]
- **Kangmun Kim** (First Lieutenant at Cyber Operations Command) Jan 2024 – Sep 2024
Topic: Web shell detection via user behavior embedding [D1]

Reference

Sangjin Lee

Professor, School of Cybersecurity, Korea University

Relationship: Ph.D. Advisor

Email: sangjin@korea.ac.kr

Seokhie Hong

Vice President, SmartM2M

Former Professor, School of Cybersecurity, Korea University

Relationship: Ph.D. Advisor

Email: shhong@korea.ac.kr

Changhee Choi

Associate Professor, Department of Cyber Defense, Sejong University

Relationship: Research Mentor (Former PI at ADD)

Email: choich@sejong.ac.kr

Jongin Lim

Professor Emeritus, School of Cybersecurity, Korea University

Former Special Presidential Advisor for Cybersecurity, Republic of Korea

Relationship: Undergraduate Advisor

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